OVERVIEW

India Agricultural Performance Dashboard

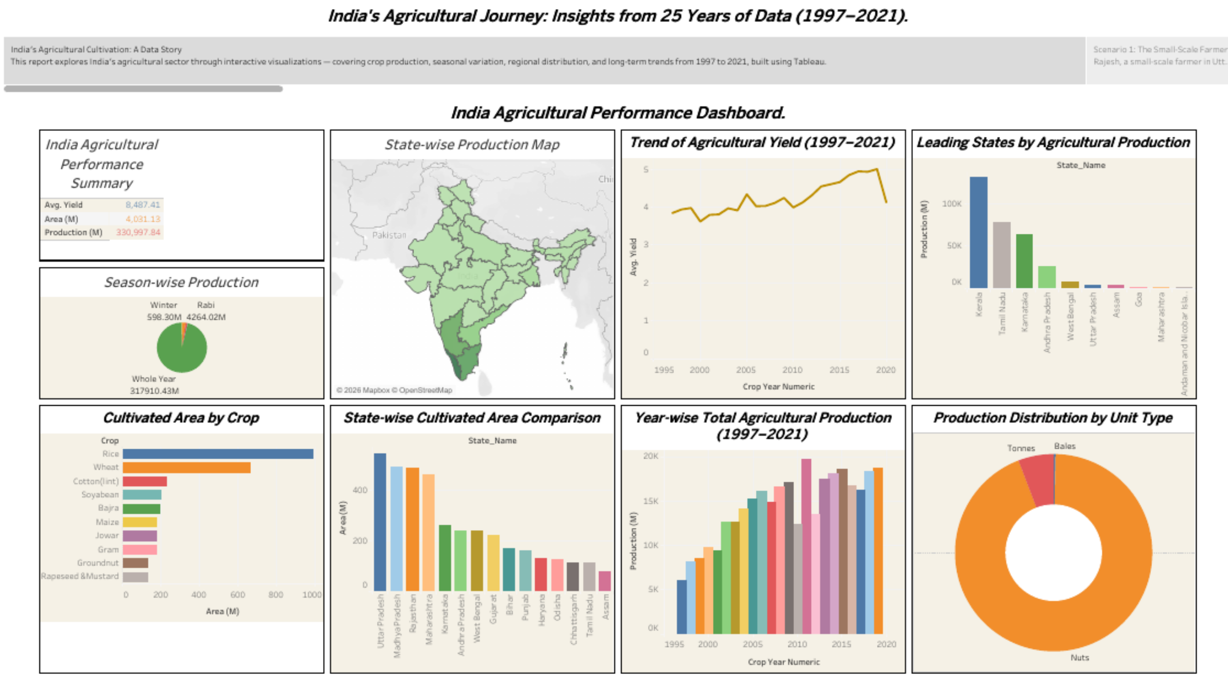

Figure 1. India Agricultural Performance Dashboard — national summary of crop production, cultivated area, yield, season and unit-type distribution across India, 1997–2021.

Key Observations

- Summary KPIs across the full 1997–2021 dataset: Avg. Yield **8,487.41**, Cultivated Area **4,031.13M**, and Total Production **330,997.84M**.
- Rice and Wheat account for the largest cultivated area, followed by Cotton (lint), Soyabean, Bajra, Maize, Jowar, Gram, Groundnut, and Rapeseed & Mustard.
- Kerala, Tamil Nadu and Karnataka lead the production ranking, while Uttar Pradesh, Madhya Pradesh and Rajasthan lead in cultivated area — the two rankings do not match.

Insight

Insight: Production leadership and cultivated-area leadership diverge because the dataset's “Production” figure blends incompatible units — coconuts are counted in “Nuts” rather than tonnage, which inflates southern coconut-producing states. Any state-to-state comparison of “output” should first be normalized by unit type, otherwise it overstates the productivity of nut-heavy states relative to grain-heavy ones.

Farmer's View — Seasonal & Crop Insights

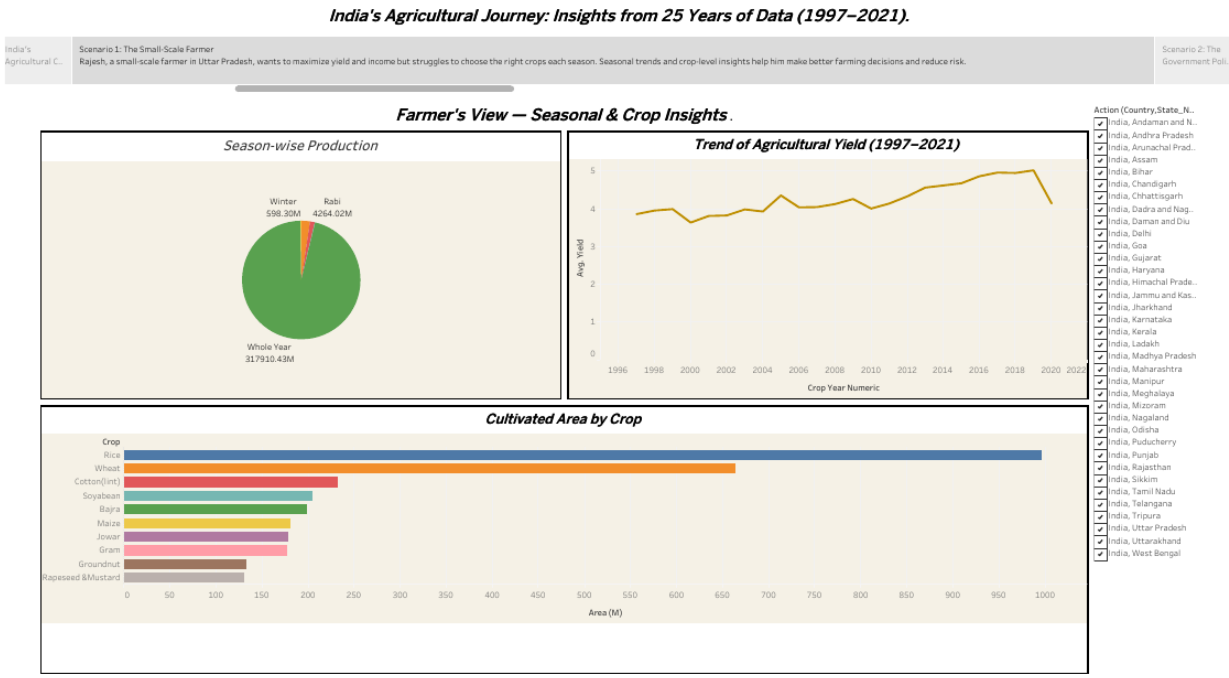

Figure 2. Farmer's View — season-wise production split and crop-level cultivated area, built to help a smallholder such as Rajesh (Uttar Pradesh) choose crops each season.

Key Observations

- Whole Year cropping dominates production (**317,910.43M**), far outweighing the Rabi (**4,264.02M**) and Winter (**598.30M**) seasons combined.
- Rice occupies the largest cultivated area, followed by Wheat and Cotton (lint) — confirming staple-crop dominance in national farming patterns.
- Average yield rose from roughly 3.5 in the late 1990s to a peak near 5 around 2018–2020, before a slight dip toward 2021.

Insight

Insight: For a smallholder like Rajesh, the seasonal split suggests that Whole-Year and Kharif-aligned staple crops (rice, wheat) historically carry the lowest seasonal risk given their dominant production volume. The recent yield dip after 2018–2020 is a signal to plan for input-cost and monsoon-driven risk rather than assume yields will keep climbing.

Policy Analyst's View — Regional Performance & Trends

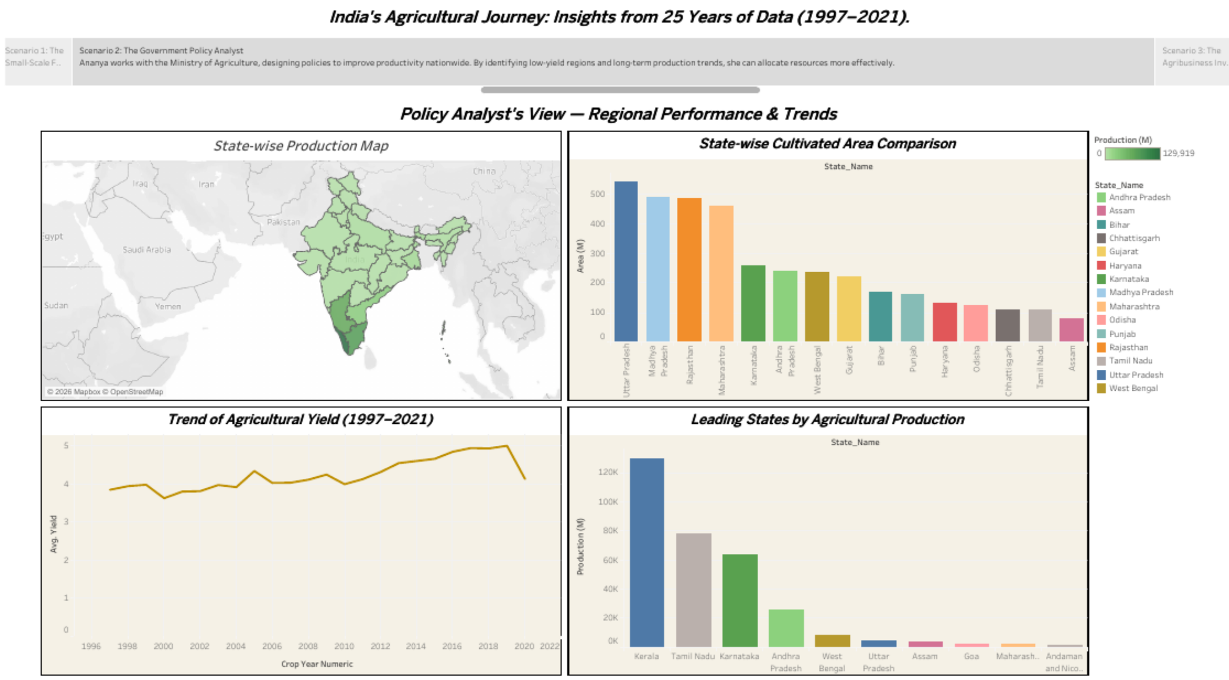

Figure 3. Policy Analyst's View — state-level cultivated area and production, ranked to support resource-allocation decisions such as those made by Ananya at the Ministry of Agriculture.

Key Observations

- Uttar Pradesh, Madhya Pradesh, Rajasthan and Maharashtra together hold the largest cultivated areas nationally.
- Kerala, Tamil Nadu and Karnataka top the production ranking (up to roughly 129,919M), despite not leading in cultivated area.
- The 25-year yield trend line shows a broadly upward path (≈3.5 to ≈5) with year-to-year volatility consistent with monsoon dependence.

Insight

Insight: The mismatch between "largest cultivated area" states and "highest production" states points to uneven productivity, not just uneven land. Ananya's team could prioritize yield-improvement programs — irrigation, MSP support, extension services — in high-area, comparatively lower-yield states such as Uttar Pradesh and Madhya Pradesh, and use rainfall-anomaly monitoring (as outlined in the enterprise architecture's Climatic Shock Factor) to explain year-to-year yield swings.

SCENARIO 3 · THE AGRIBUSINESS INVESTOR

Investor's View — Growth & Opportunity Insights

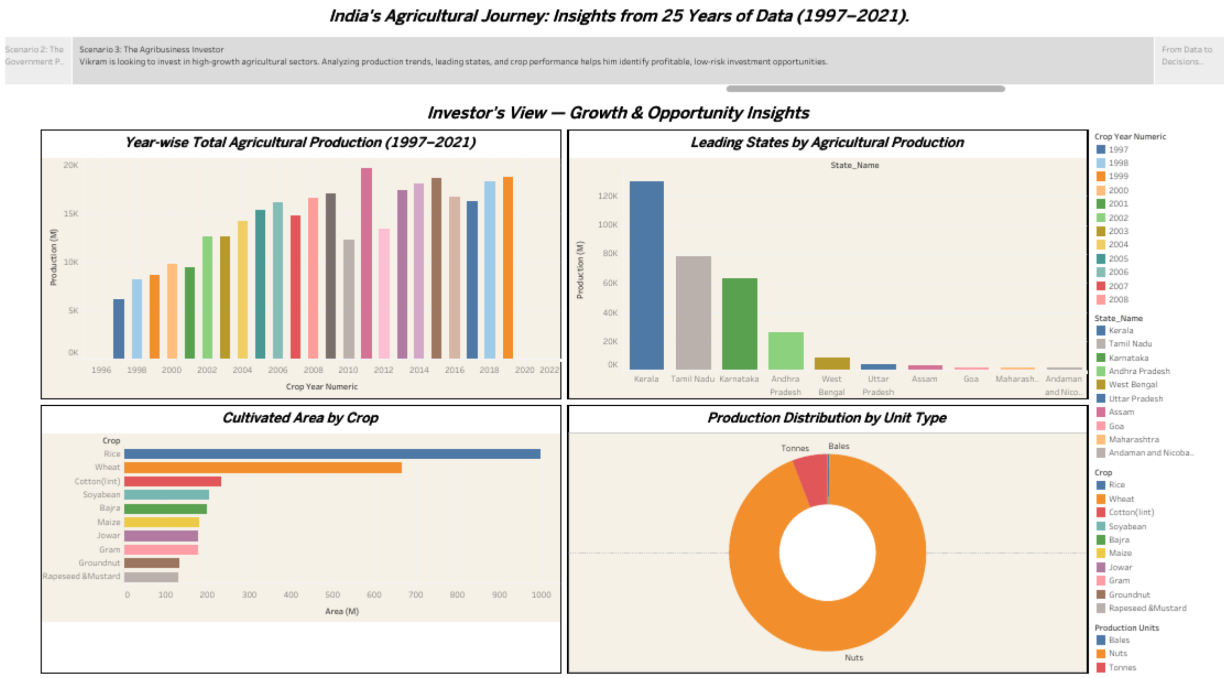

Figure 4. Investor's View — year-wise production growth, top-producing states and unit-type composition, framed for screening decisions such as Vikram's.

Key Observations

- Total production grew steadily from the late 1990s, accelerated through the 2000s, and peaked in the mid-to-late 2010s before a slight decline toward 2021.
- “Nuts” dominate the Production Distribution by Unit Type chart, with Tonnes and Bales making up a smaller remainder.
- Kerala, Tamil Nadu, Karnataka, Andhra Pradesh and West Bengal form the top five states by reported production volume.

Insight

Insight: Vikram should treat raw “Production” totals cautiously, since they mix incompatible units (Nuts, Tonnes, Bales). A fair growth-and-risk comparison across states requires normalizing by unit type — or evaluating tonnage-based staple crops separately from coconut/nut-based output — before treating any state's production volume as a direct investment signal.

From Data to Decisions

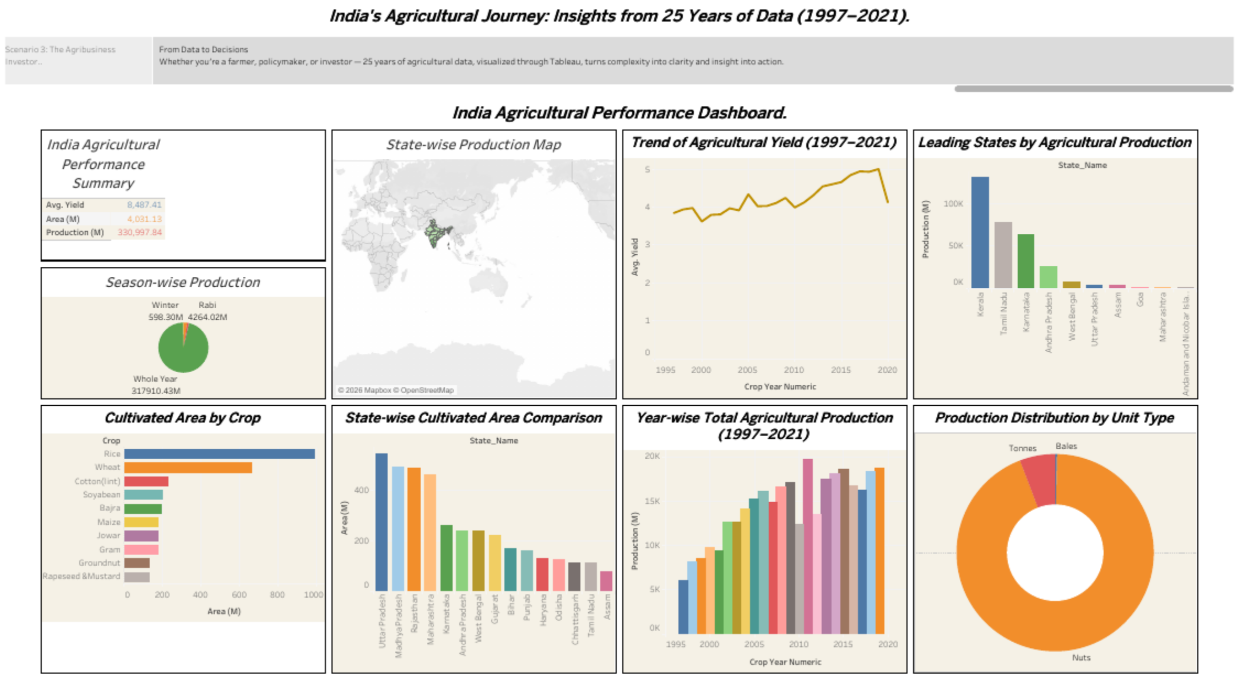

Figure 5. India Agricultural Performance Dashboard (closing view) — the same national summary revisited after the three role-based scenarios.

Key Observations

- The same core KPIs (Avg. Yield 8,487.41; Area 4,031.13M; Production 330,997.84M) anchor all three stakeholder views, confirming a single consistent Tableau data source.
- Across all three scenarios, recurring themes are staple-crop (rice/wheat) dominance, monsoon-linked yield volatility, and a production-vs-cultivated-area mismatch across states.
- The storyboard shows one workbook can serve a farmer, a policymaker and an investor by re-framing the same underlying data around each user's decision needs.

Insight

Insight: The project's core value is turning one static 1997–2021 dataset into three decision-ready lenses. Since the data does not extend past 2021 or include live weather or soil variables, the logical next step — per the accompanying enterprise architecture roadmap — is integrating near-real-time ingestion (Meteorological Networks, Wholesale Marketplace pricing) so these insights stay current for real-world decisions.